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 UNIVERSITY OF TECHNOLOGY FACULTY OF CSE	FINAL EXAM		Semester / Academic year		3	2022-2023
			Date		04/08/2023	
	Course title	Programing Fundamentals				
	Course ID	CO1027				
	Duration	70 mins	Question sheet code		2231	
Notes: - Students are not allowed to use course materials. - Submit this question sheet together with the answer sheet. - Choose the best answer (only 1) for each question.						

- (L.O.2) Which of the following is a User-defined data type?
A. typedef int Boolean; B. typedef enum {Mon, Tue, Wed, Thu, Fri} Workdays;
C. struct {char name[10], int age}; D. All of the mentioned;
- (L.O.2) Which of the following statements is correct when a class is inherited publicly?
A. Public members of the base class become protected members of the derived class. B. Public members of the base class become private members of the derived class.
C. Private members of the base class become protected members of the derived class. D. Public members of the base class become public members of the derived class.
- (L.O.2) Which of the following declaration is illegal?
A. char *str = "string Contents"; B. char str[] = "string Contents";
C. char str[20] = "string Contents"; D. char[] str = "string Contents";
- (L.O.3) To ensure that every object in the array receives a destructor call, always delete memory allocated as an array with the operator:
A. destructor B. New[] C. delete[] D. Free[]
- (L.O.1) What will be the values of x , m and n after the execution of the following statements?

```
1 int x, m, n;
2 m = 10; n = 15;
3 x = ++m + n++;
```

A. x=25, m=10, n=15 B. x=26, m=11, n=16 C. x=27, m=11, n=16 D. x=27, m=10, n=15
- (L.O.2) Within a switch statement
A. Continue can be used but Break cannot be used B. Continue cannot be used but Break can be used
C. Both Continue and Break can be used D. Neither Continue nor Break can be used
- (L.O.2) What will be the output after the execution of the following statements?

```
1 #include<stdio.h>
2 int main() {
3     int s=0;
4     while(s++<10) {
5         if(s>5 && s<10)
6             continue;
7         printf("\n%d\t", s); }
8     return 1;}
```

A. 1 2 3 4 5 10 B. 2 3 4 5 10 C. 2 3 4 10 D. 2 3 10

8. (L.O.4) For the assignment statement $a=b$; Which statement is true?

- A. A check is done to compare the values of a and b.
- B. The value of b is assigned to variable a and any further changes in the program on variable b will also change the value of variable a.
- C. The value of b is assigned to variable a and any further changes in the program on variable b will not change the value of variable a.
- D. The value of b is assigned to variable b and any further changes in the program on variable a will not change the value of variable b.

9. (L.O.2) Observe the following block of code and determine what happens when $x = 2$?

```
1 #include<stdio.h>
2 int main() {
3     int x=2;
4     switch(x) {
5         case 1: printf("x is 1"); break;
6         case 2:
7         case 3: printf("x is 3"); break;
8         default: printf("x is not within the range");
9     }
10    return 1; }
```

- A. Program jumps to the end of switch statement since there is nothing to do for $x=2$.
- B. The code inside default will run since there is no task for $x=2$.
- C. Will display x is 3, and then come outside the switch statement.
- D. None of the other answers is correct

10. (L.O.3) Which of the following is true?

- A. All objects of a class share all data members of the class.
- B. Objects of a class do not share non-static members. Every object has its own copy.
- C. Objects of a class do not share codes of non-static.
- D. None of the other answers is correct

11. (L.O.1) Find the output of the following program:

```
1 #include<stdio.h>
2 int main() {
3     int x=5;
4     if(x=0)
5         printf(" = am in Zero");
6     else
7         printf(" = am in :ero");
8     return 1; }
```

- A. = am in Zero
- B. = am in :ero
- C. Compiler error
- D. None of the other answers is correct

12. (L.O.2) what is the mechanism of giving special meaning to an operator

- A. Operator overloading
- B. Function overloading
- C. Constructor overloading
- D. Virtual function

13. (L.O.2) Find the output of the following program

```
1 #include<iostream>
2 using namespace std;
3 int main() {
4     int i, a[2]={10,20};
5     for(i=0; i<=1; i++)
6     {
7         cout<< *(a+1)<< " "<< *(i+a)<< " ";
8     }
9     return 1;
10 }
```

- A. 20 20 20 20
- B. 10 10 10 10
- C. 20 10 20 10
- D. 20 10 20 20

14. (L.O.2) Given the following program:

```

1 class par{
2     protected:
3     void protect(){};
4 };
5 class chil: public par{
6     void protect(){par::protect();}
7 };
8 int main(){
9     par a;
10    a.protect();
11    return 1;
12 }

```

Assume that all the necessary libraries are included. What is the result of this program?

- A. Error in line 10
- B. Program execute without any problem
- C. Error in lines 6 and 10
- D. Error in line 6

15. (L.O.3) Given the following program:

```

1 int sum(int a,int b){return a+b;}
2 int sum(double a,double b){return a+b;}
3 int main(){
4     cout<<sum(3.5,5);
5     return 1;
6 }

```

Assume that all the necessary libraries are included, What is the output on display of this program?

- A. 8
- B. 8.5
- C. 8.50000
- D. None of the other answers is correct

16. (L.O.3) Given the following program:

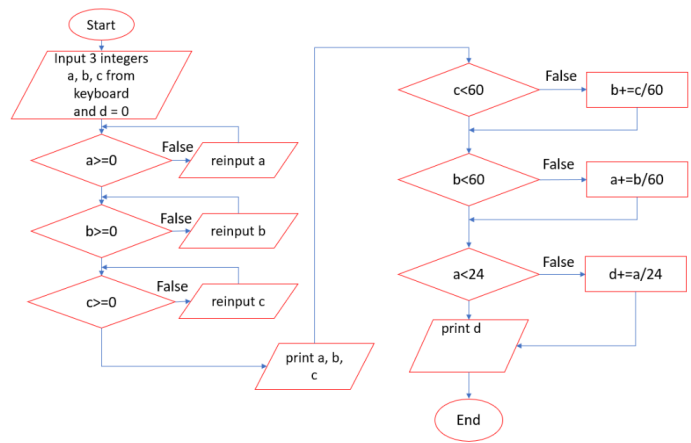
```

1 int findMax(int A[],int n=5){
2     static int max=0;
3     if(n<1) return max;
4     max=A[n]>max?A[n]:max;
5     return findMax(A,n-1);
6 }
7 int main(){
8     int A[]={6,7,8,9,10};
9     int B[]={1,2,3,4,5};
10    cout<<findMax(A)<<" "<<findMax(B);
11    return 1;
12 }

```

Assume that all the necessary libraries are included. What is the output display of this program?

- A. 10 10
- B. 10 5
- C. 0 0
- D. 1 6



17. (L.O.2) What is the output when the input is:
23 60 -5 -2 -3 0

- A. 23 60 0 1
- B. 23 60 0 0
- C. 0 60 23 0
- D. 23 60 -5 0

18. (L.O.2) What is the output when the input is:
-3 -4 0 24 -5 25

- A. 24 25 0 1
- B. 0 25 24 0
- C. 0 24 25 0
- D. 25 24 0 1

19. (L.O.2) What is the output when the input is:
-3 50 600 23 -5 25

- A. 23 50 600 1
- B. 50 600 23 2
- C. 50 600 25 2
- D. Error

20. (L.O.3) Given the following function:

```

1 void printArray(int A[],int n){
2     if(n){
3         cout<<A[n-1]<<" ";
4         printArray(A,n-1);
5         cout<<A[n-1]<<" ";
6     }
7 }

```

What is the output on display when input A=1,2,3,4,5 and n=5 to this function?

- A. 1 2 3 4 5 5 4 3 2 1
- B. 5 4 3 2 1 1 2 3 4 5
- C. 1 2 3 4 5 1 2 3 4 5
- D. 5 4 3 2 1 5 4 3 2 1

21. (L.O.1) Knowing that a char data is 1byte: Let char a=126. What is the value of char a after the calculation: a=(a+5)%5;

- A. char(1)
- B. char(2)
- C. char(0)
- D. char(131)

22. (L.O.3) Given a dynamic array pointed to by pointer A: {1,3,5,7,9,11,13,15} After the following calculation: pointer B=A+5; A=B+5; B=A-4;. Which element of the array is currently pointed by pointer B?

- A. 9
- B. 11
- C. 13
- D. None of the other answers is correct

For questions 17–19, consider the following flow chart:

23. (L.O.3) Given the following function:

```
1 void func(int *a, int *b){
2     int *c=a;
3     a=b;
4     b=a;
5 }
6 int main(){
7     int *a=new int(5);
8     int *b=new int(7);
9     func(a,b);
10    return 1;
11 }
```

Assume that all the necessary libraries are included, What is the value of the integers pointed to by a and b respectively when input int *a=new int(5) and int *b= new int(7) to this function?

- A. 5 7 B. 7 5
C. 5 5 D. 7 7

24. (L.O.4) Given the following program:

```
1 class human{
2     void makeSound() {cout<<"speak";}
3 };
4 class singer:public human{
5     void makeSound() {cout<<"sing a song";}
6 };
7 int main(){
8     human* pS=new singer();
9     pS.makeSound();
10    return 1;
11 }
```

Assume that all the necessary libraries are included, what is the output on display of this program?

- A. sing a song B. speak
C. Error because of line 8
D. None of the other answers is correct

25. (L.O.3) Given an integer pointer A, A pointed to an array of 10 elements. Knowing that A=0xe927e1, what is the value of A+4;

- A. 0xe927f1 B. 0xe927e5
C. 0xe927e17 D. 0xe92801

26. (L.O.3) Given the following program:

```
1 int main() {
2     int B[]={1,3,5,7,9};
3     int *A=B;
4     for(int i=0;i<5;i++){
5         cout<<A;
6         A++;
7     }
8     return 1;
9 }
```

Assume that all the necessary libraries are included. What is the output display of this program?

- A. None of the other answers is correct
B. 1 3 5 7 9 C. Error
D. 9 7 5 3 1

27. (L.O.3) Given the following function:

```
1 int findMin(int A[], int n){
2     int min=0;
3     for(int i=0;i<n;i++){
4         if(A[i]>min){
5             min=A[i];
6         }
7     }
8     return min;
9 }
```

When input array A={-5,-3,-2,-10} and n=4, what is the function's output?

- A. -10 B. -2
C. -5
D. None of the other answers is correct

28. (L.O.3) Given the following program:

```
1 int sum(int a, int b){return a+b;}
2 int main() {
3     cout<<sum(3.5,5);
4     return 1;
5 }
```

Assume that all the necessary libraries are included, What is the output on display of this program?

- A. 8 B. 8.5
C. 8.50000
D. None of the other answers is correct

29. (L.O.3) Given char a[]="1 2 3 4 5 \0 5 4 3 2 1"; What is the value of strlen(a)?

- A. 10 B. 20
C. 11 D. 22

30. (L.O.4) Given the following class:

```
1 class stringMat{
2     string** Mat;
3     string*& operator [] (int i){return Mat[i];}
4 };
```

Given a stringMat with Mat value of {"slot 1", "slot 2"}, {"slot 3", "slot 4"} What is the value of Mat[1][1] and Mat[0][1] respectively?

- A. slot 3 and slot 4
B. slot 4 and slot 3
C. slot 4 and slot 2
D. Error because operator[] can only access string** and not string*